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27 October 2026

To the Editors,
Expert Systems with Applications

We submit for your consideration the manuscript *Deep Reinforcement Learning for the Interval Job Shop Scheduling Problem: A Comparison with Genetic Programming Hyper-Heuristics across Inference Budgets*, as a research paper.

The interval job shop scheduling problem models each uncertain processing time as an interval between known bounds, and the literature solves it with metaheuristics that search anew on every instance. The manuscript makes two contributions. The first is a constructive deep reinforcement learning policy for the problem. Its features are dimensionless and its encoder is permutation invariant, so a single network trained on four 20×15 instances schedules any other size without retraining or architectural change. It improves on every hand-crafted dispatching rule on all seventy instances of the benchmark, six of whose seven size classes it never saw. The second is a controlled comparison between the two learned constructive paradigms available for the problem: thirty trained artifacts of each family, evaluated in one harness, under one selection criterion and at matched inference budgets. The comparison finds no dominance. Which learner leads is decided by the inference budget and, on instances neither family has seen, by the size class.

The work fits the journal because it gives practical guidance for building intelligent scheduling systems under uncertainty. It states when a compact, readable evolved rule is the better choice, when a trained policy's sampling distribution earns its higher cost, and what each costs to train and to run. The journal has published deep reinforcement learning for job shop scheduling evaluated on the same benchmark families, and the manuscript builds on that line of work.

We confirm that the manuscript is original, that it has not been published elsewhere, and that it is not under consideration by another journal. We disclose one related work: a companion study by the first author on genetic programming hyper-heuristics for this problem is under review in another journal and available as a preprint (SSRN, doi:10.2139/ssrn.7214286). The present manuscript is distinct in both contributions above. That study enters it only as the source of the thirty evolved rules it compares against.

The instances, source code, experiment records and trained checkpoints supporting the study are openly available in Zenodo at doi:10.5281/zenodo.21970431. A statement disclosing the use of generative AI in the preparation of the manuscript appears directly before the references, as the journal's policy requires. In keeping with the double-anonymized review process, author identity appears only in the title page and in this letter.

Thank you for considering our work.

Yours sincerely,

Hernán Díaz Rodríguez
Corresponding author, on behalf of both authors